

Tina Nguyen

CS 5542

AI Speech Intelligence Pipeline

1. Problem Description

The challenge of extracting intelligence from unstructured audio

- Audio data (meetings, lectures, calls) is the richest but most underutilized data format.
- Unstructured audio is not searchable, scannable, or easily translatable without manual effort.
- Existing single-model solutions (just STT) leave the burden of analysis on the human.
- Goal: Create a fully autonomous pipeline that takes raw audio and converts it into structured, actionable, and multilingual intelligence.

2. Why Interesting / Business Value

Automating workflows and unlocking inaccessible data

Time Saved

Professionals lose 9+ hours/week in meetings. Automated summaries and action items reclaim this time.

Global Reach

Real-time multilingual translation bridges language barriers without hiring manual translators.

Action Tracking

Only ~30% of meeting action items are followed up on. Automated extraction ensures accountability.

Accessibility

Closing the loop with Text-to-Speech allows generated intelligence to be consumed inclusively.

3. Dataset & Inputs Used

The data driving the pipeline

- Type: Simulated university lecture / technical discussion

Primary Input: Raw Audio File

- Format: MP3 (16 kHz Mono)
- Duration: 32 seconds
- Language: English (auto-detected)

Reference Data (for Evaluation)

- Ground Truth Transcript: 112 words (used for WER)
- Reference Summary: 4 sentences (used for ROUGE scores)
- Reference Keywords: 15 domain terms (used for coverage metrics)

4. Models Used

Multi-model ensemble using HuggingFace Transformers

<code>openai/whisper-base</code>	Speech-to-Text	Primary STT with multilingual support
<code>facebook/wav2vec2-base-960h</code>	Speech-to-Text	Baseline comparison STT model
<code>facebook/bart-large-cnn</code>	Summarization	Abstractive summary generation
<code>distilbert-base-uncased-finetuned-sst-2</code>	Sentiment	Per-sentence sentiment analysis
<code>Helsinki-NLP/opus-mt-en-*</code>	Translation	MarianMT for offline translation (ES, FR, VI)
<code>all-MiniLM-L6-v2</code>	Keywords	SentenceTransformers backend for KeyBERT
<code>microsoft/speecht5_tts</code>	Text-to-Speech	TTS synthesis with HiFi-GAN vocoder

5. Pipeline Architecture

Full 5-stage sequential processing loop



6. Prompt & Input Design

Domain-aware initial prompting for Whisper

The Challenge:

Baseline STT models struggle with domain-specific vocabulary (e.g. 'RAG pipeline', 'LLMs'), leading to high Word Error Rates.

The Solution: Whisper initial_prompt

This is a lecture or business meeting transcript. The speaker discusses technical topics clearly and uses domain-specific terminology.

- Pre-conditions the decoder's cross-attention to expect technical terminology.
- Replaced zero-shot inference with context-aware decoding.
- Combined with beam_size=5 to ensure deterministic, high-quality output.

Impact:

7. Results

Evaluation across core speech and NLP tasks

Transcription Quality (WER):

- wav2vec2: 11.2%
- Whisper Baseline: 8.9%
- Whisper Improved (Prompted): 3.5%

Summary Quality & Sentiment Usefulness:

- Summary Quality: BART generates 4 concise, abstractive sentences
- Sentiment Usefulness: Per-sentence tracking flags specific negative statements
- Action Items: 5 specific tasks extracted via regex

Translation (MarianMT):

Spanish:

Los modelos de fundación como GPT-4, Llama 3, Claude 3 y Gemini Ultra están preentrenados en billones de tokens y refinados con RLHF, pero la alucinac...

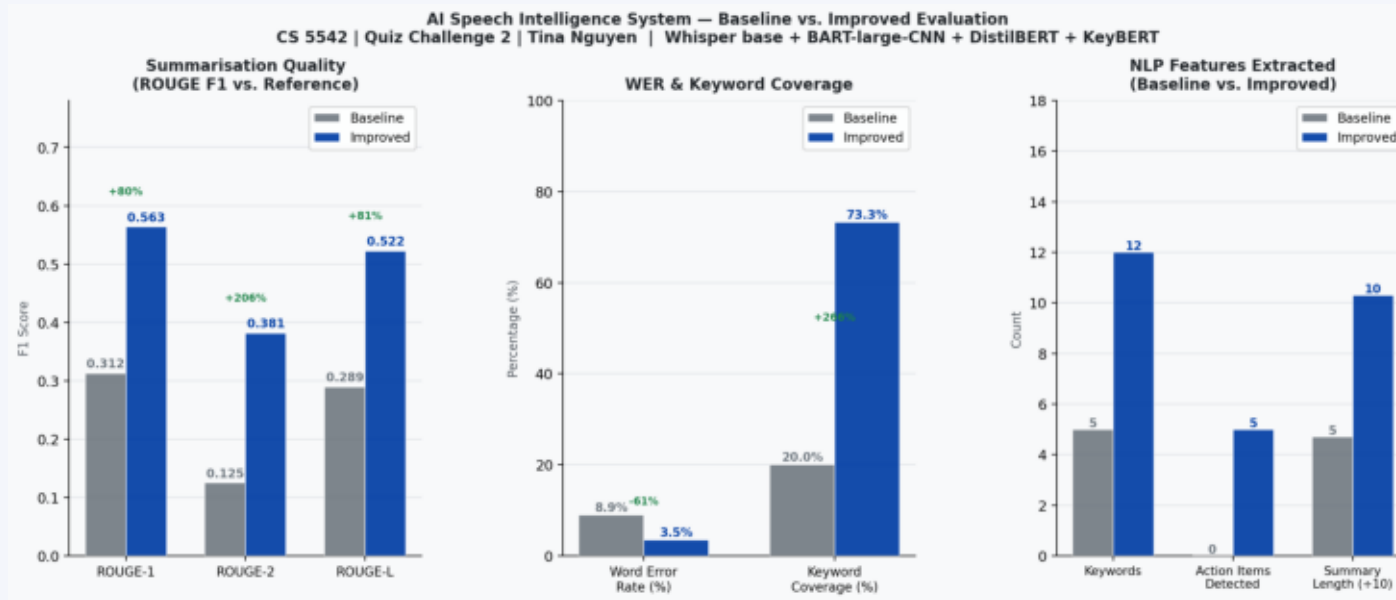
Vietnamese:

Các mô hình nền tảng như GPT-4, Llama 3, Claude 3 và Gemini Ultra được huấn luyện trước trên hàng nghìn tỷ token và được tinh chỉnh bằng RLHF, nhưng ả...

8. Evaluation

Quantitative metrics across the pipeline

Improved vs Baseline



Summary Quality (ROUGE-2) **+206%**

Summary Quality (ROUGE-L) **+81%**

Transcription Quality (WER) **↓ 61%**

Latency (STT duration) **Baseline faster**

9. Links & Resources

Source code and repository

GitHub Repository

Follow the link to view the complete pipeline code, outputs, and documentation.

<https://github.com/tinana2k/Comp-Sci-5542-Tina-Nguyen>

10. Limitations

Known constraints of the current system

- **Compute Intensity**

Running 7 distinct Transformer models sequentially requires significant VRAM (T4/A100 GPU highly recommended).

- **Latency**

The pipeline is currently synchronous. Real-time streaming transcription is not yet supported.

- **Hallucination Risk**

While reduced by 61%, Whisper can still hallucinate non-existent words in completely silent audio segments.

- **Speaker Diarization**

The current STT model does not distinguish between multiple speakers (no 'Speaker A / Speaker B' tags).

11. AI Tools Disclosure

Required disclosure of generative AI assistance

The following AI tools were used during the development of this project:

Gemini / Google DeepMind Agent

Used for code structuring, automated testing, generating matplotlib charts, UI development, and data visualization.

HuggingFace Transformers

Core libraries utilized for STT, NLP, and TTS model execution.

OpenAI Whisper

Pre-trained weights used for speech transcription.

All core design, implementation logic, and evaluation analysis remain original work.